

1. Consider the discrete-time signal $\mathbf{X} = \{1, 2, 3, 4, 5\}$ and the linear filter with impulse response $\mathbf{a} = \{\frac{1}{2}, 1, \frac{1}{2}\}$.
 - (a) Compute the filtered series $\mathbf{Y}$ using zero-padding for values outside the range of the signal.
 - (b) Compute the frequency response of the filter.
2. Consider a moving average smoothing filter of window size $L = 2m + 1$, where m is an integer.
 - (a) Write the impulse response $a(t)$.
 - (b) Compute the frequency response $A(\omega)$.
 - (c) Compute the power response.
 - (d) What are the disadvantages of using this filter as low-pass filter?
3. Consider the signal $x_t = 2 \cos(2\pi \cdot 10t) + \cos(2\pi \cdot 30t)$ sampled at 100 Hz for 1 second.
 - (a) Sketch the DFT magnitude spectrum and the periodogram of the signal, including both positive and negative frequencies.
 - (b) Imagine you only want to retain the 10 Hz component, what type of linear filter would you use? Sketch the frequency response of the filter.
 - (c) Now, consider the signal shown in fig. 1, along its discrete Fourier transform (DFT) magnitude spectrum. The signal is sampled at 800 Hz, and no aliasing is present.

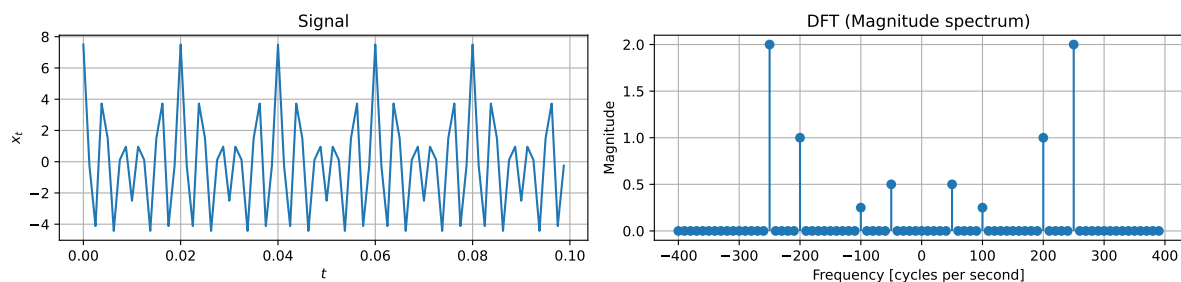


Figure 1: Signal and its DFT magnitude spectrum.

Write the mathematical representation of the signal in terms of its frequency components.

- (d) Suppose the signal in fig. 1 is sampled at a rate of 450 Hz. Sketch the updated DFT magnitude spectrum and explain the changes. Determine the minimum sampling frequency required to avoid aliasing for this signal.